

🏠 Executive Summary: Mobile Edge Computing (MEC) vs. Traditional Cloud Computing

1. The Core Problem: Why Traditional Cloud Falls Short for 5G

In **Traditional Cloud Computing**, computing and storage are centralized in massive, remote data centers located far from the end users.

The Communication Path:

$$\text{User} \rightarrow \text{5G Base Station} \rightarrow \text{Mobile Core Network} \rightarrow \text{Internet / WAN} \rightarrow \text{Cloud Data Center}$$

The Consequence: Because data packets must cross multiple network hops and traverse the wide-area transit network (WAN), the round-trip latency is relatively high (**~80–180 ms** in real-world scenarios, demonstrated as **~120 ms Simulated RTT**).

Limitation: Centralized clouds struggle to meet the strict real-time requirements of next-generation applications such as autonomous driving, real-time video analytics, AR/VR, and industrial automation.

**2. The Solution: Mobile Edge Computing (MEC)

Mobile Edge Computing (MEC) fundamentally solves this problem by **bringing computing, storage, and AI processing resources closer to mobile users**—physically co-located at or near the **5G Base Station (gNodeB)**.

The Communication Path:

$$\text{User} \rightarrow \text{5G Base Station} \rightarrow \text{MEC Host (Co-located Edge)} \rightarrow \text{Immediate Result to User}$$

The Advantage: Data terminates directly at the local edge host without crossing the Mobile Core or the public Internet. This cuts network hops down to **1 hop**, drastically reducing simulated latency to **~15 ms**.

Bandwidth Savings: High-bandwidth raw data (such as 4K video feeds) does not congest the telecom core network.

**3. Architectural Comparison Table

| Feature |  Traditional Cloud Computing |  Mobile Edge Computing (MEC) |

| :--- | :--- | :--- |

| **Compute Placement** | Centralized in remote hyperscale datacenters | Decentralized; co-located at the 5G Base Station |

| **Network Hops** | **4 Hops** (Base Station $\rightarrow$ Core $\rightarrow$ WAN $\rightarrow$ Cloud) | **1 Hop** (Direct Base Station to MEC Host link) |

| **Simulated RTT** | **~120 ms** (Higher delay due to WAN transit) | **~15 ms** (Near real-time response) |

| **Core Network Load** | High (All raw data traverses backhaul) | Very Low (Data filtered & processed locally) |

| **Mobility Extension** | None (Static remote facility) | **UAV-MEC** (Aerial edge nodes on drones) |

| **Best Suited For** | Big data storage, heavy model training, global services | Real-time AI inference, video analytics, AR/VR, IoT |

4. How to Demonstrate This in the Simulation

1. **Switch to Traditional Cloud Mode ([TRADITIONAL CLOUD]):**

- * Point out that **all MEC nodes disappear**.
- * Show the animated packet traveling the long vertical path: User $\rightarrow$ Tower $\rightarrow$ Core $\rightarrow$ WAN $\rightarrow$ Cloud.
- * Highlight the **“CLOUD PROCESSING”** delay before the result returns across the WAN.

2. **Switch to MEC Mode ([MEC]):**

- * Show the **“MEC HOST”** co-located right next to the 5G Base Station with the green **“HIGH-SPEED EDGE LINK”**.
- * Point out the edge modules: **CPU, GPU, Edge Apps, Cache, and AI Services**.
- * Demonstrate how tasks finish at the edge immediately, with only periodic model syncs sent to the cloud.
- * Toggle **“[ UAV-MEC ON]”** to explain how aerial drones act as mobile edge nodes for disaster relief or temporary events.

3. **Open Compare Mode ([ COMPARE]):**

* Click **[ RUN SAME TASK: AI VIDEO ANALYTICS]**.

* The audience will see the **MEC workload finish in ~1.2 seconds**, while the **Cloud packet is still traveling through the WAN backbone**.

5. Key Educational Takeaway to Conclude Your Presentation

> **"MEC does not replace cloud computing. Instead, MEC and the Cloud collaborate: latency-critical tasks are processed immediately at the edge by the MEC Host, while large-scale analytics, centralized storage, and foundational model training remain in the Cloud."**

PPT: Mobile Edge Computing (MEC)

Slide 1 — Title

Mobile Edge Computing (MEC)

Bringing Computing Closer to Mobile Users

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B.Tech Information Technology

Guru Ghasidas Vishwavidyalaya

Visual: Use your MEC/UAV network animation as the background or a simplified architecture.

 **Say:**

“Today I am going to explain Mobile Edge Computing, or MEC, a technology that brings computing resources closer to mobile users instead of relying only on distant cloud data centers.”

Slide 2 — Why Do We Need MEC?

The Problem with Traditional Cloud Computing

 Mobile User



 Base Station



 Core Network



 Distant Cloud



 Processing



 Result

Problems

- Data travels through multiple network layers
- Higher communication latency
- Network congestion
- Large amount of data transferred to the cloud
- Not ideal for real-time applications

 Say:

“In traditional cloud computing, a mobile device often sends its data to a centralized cloud for processing. When the application requires a very fast response, the communication path and network conditions can become a limitation.”

Slide 3 — What is Mobile Edge Computing?

MEC = Computing at the Network Edge

MEC places computing, storage and application services closer to mobile users.

Traditional

User → Network → Distant Cloud

MEC

User → Nearby Edge → Result

|

└─ Cloud when required

Key Idea

Move computing closer to where data is generated.

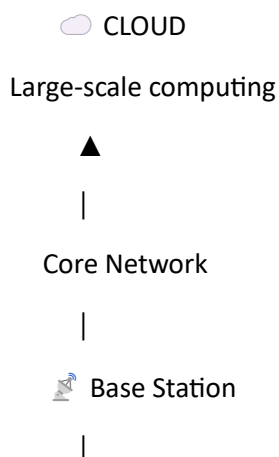
 Say:

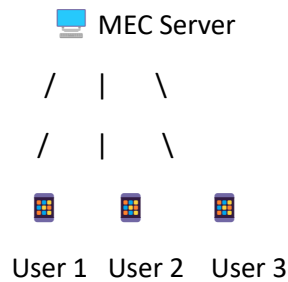
“Mobile Edge Computing solves this problem by placing computing resources at or near the network edge. Instead of sending every piece of data to a distant cloud, some processing can happen closer to the user.”

Slide 4 — MEC Architecture

How MEC Works

Use a diagram like:





Main Components

Mobile Users

- Generate data
- Use MEC applications

Base Station

- Provides wireless connectivity

MEC Server

- Provides nearby computing/storage
- Runs edge applications

Cloud

- Provides large-scale computing and storage

Say:

“The main components are mobile users, the wireless network and base station, the MEC server, and the cloud. The MEC server sits closer to users, while the cloud provides much larger centralized resources.”

Slide 5 — How MEC Works

From Data Generation to Result

- ①  Data Generated
↓
- ②  Wireless Network
↓
- ③  MEC Processing
↓
- ④  Application/AI
↓

⑤ 📱 Result Returned

Example: AI Video Analytics

📷 Camera



Video Data



💻 MEC



🤖 AI Analysis



"Person / Vehicle Detected"

🎤 **Say:**

"For example, a mobile camera can send video data to a nearby MEC server. The edge server can perform AI-based video analysis and return the result quickly, instead of sending the entire video stream to a distant cloud."

Slide 6 — Why is MEC Important?

Advantages of MEC

⚡ Low Latency

Processing happens closer to users.

📊 Reduced Network Traffic

Less data needs to travel to the central cloud.

🚀 Faster Response

Useful for time-sensitive applications.

🔒 Better Data Locality

Some data can be processed closer to where it is generated.

📈 Scalability

Computing resources can be distributed across multiple edge locations.

🤝 Cloud + Edge Collaboration

MEC complements rather than completely replaces cloud computing.

🎤 **Say:**

“The major benefit of MEC is that it can reduce the communication distance between the user and computing resources. This makes it particularly useful for applications where response time is important.”

Slide 7 — MEC Applications

Where is MEC Used?

Application	Why MEC?
 AI Inference	Fast processing
 AR/VR	Very low response time
 Connected Vehicles	Real-time decisions
 Video Analytics	Local processing of large video streams
 Smart Cities	Process IoT data near the source
 Cloud/Edge Gaming	Reduce interaction delay
 Industrial IoT	Real-time monitoring

Say:

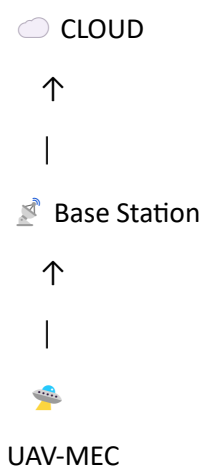
“MEC can support many applications including AI inference, AR and VR, connected vehicles, video analytics, smart cities, gaming and industrial IoT.”

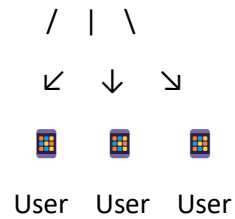
Slide 8 — UAV-Assisted MEC

This is where you can make your presentation **unique**.

UAV-MEC

UAVs can carry computing resources and act as mobile edge nodes.





Why UAV-MEC?

- Edge computing can be deployed where needed
- Useful in disaster/emergency situations
- Can support temporary high-demand areas
- Can provide connectivity and computing support

Say:

“An interesting extension of MEC is UAV-assisted MEC. Here, drones can carry computing resources and act as mobile edge nodes. This can be useful in disaster areas, temporary events or locations where fixed infrastructure is insufficient.”

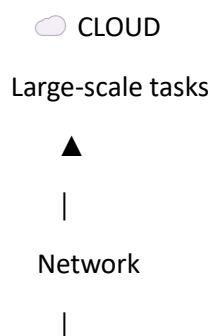
Slide 9 — MEC vs Cloud

MEC and Cloud Work Together

MEC	Cloud
Close to users	Usually centralized
Lower communication latency	More distant
Limited resources	Very large resources
Real-time applications	Large-scale processing
Local/edge processing	Global processing & storage

Important

MEC does NOT replace Cloud Computing.





Low-latency processing



“MEC is not a replacement for cloud computing. They can work together. MEC is useful for latency-sensitive processing near users, while the cloud provides large-scale computing and storage.”

Slide 10 — Interactive MEC Simulation

This is where you demonstrate your webpage.

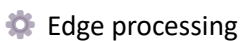
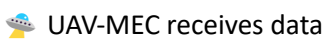
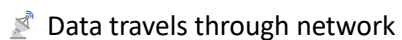
Live MEC Network Simulation

Your webpage shows:



Animation

START SIMULATION





● Result generated



📱 Result returned to users



☁ Cloud handles larger/global processing

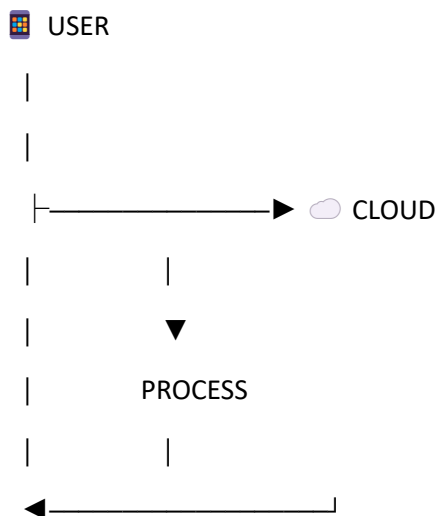
🎤 **Say:**

“To demonstrate this concept, I created an interactive MEC simulation. Here, mobile users generate different types of tasks. Data travels through the network to nearby MEC nodes, where processing occurs. The results are then returned to the users, while the cloud can provide additional large-scale resources.”

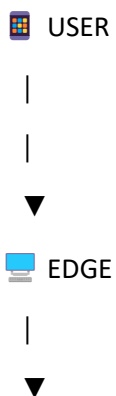
Slide 11 — MEC vs Traditional Cloud

The Core Difference

Traditional Cloud



MEC



PROCESS

|



RESULT

Key Message

MEC brings computing closer to the user.



Say:

“So the fundamental difference is simple: traditional cloud computing generally places computation in centralized data centers, while MEC places computing resources closer to the users at the network edge.”

Slide 12 — Conclusion

Conclusion

MEC enables:

-  **Lower latency**
-  **Reduced network traffic**
-  **Faster response**
-  **Real-time AI applications**
-  **Better mobile application performance**

Final Message

“Instead of always moving data to computing, MEC brings computing closer to the data.”



Final line:

“In short, Mobile Edge Computing brings computing closer to where data is generated, making it an important technology for the next generation of mobile, AI, IoT and real-time applications.”

Recommended 5-Minute Version

You **should NOT present all 12 slides in detail**. For a 5-minute class presentation, I'd actually reduce it to **8 slides**:

Slide	Topic	Time
1	Title	15 sec
2	Problem with Cloud	35 sec
3	What is MEC?	40 sec

Slide	Topic	Time
4	MEC Architecture	50 sec
5	How MEC Works	40 sec
6	Advantages + Applications	45 sec
7	UAV-MEC	35 sec
8	Live Web Simulation + Conclusion	80 sec

Total: ~5 minutes

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